

SPECTROSCOPIC VERIFICATION OF BARIUM DWARF CANDIDATES: THE ANALYSIS OF HD 8270, HD 13551, AND HD 22589¹

C. B. PEREIRA

Observatório Nacional/MCT, Rua General José Cristino 77, 20921-400 Rio de Janeiro, Brazil; claudio@on.br

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ABSTRACT

This work presents the abundance patterns of three barium dwarf candidates, HD 8270, HD 13551, and HD 22589, based on high-resolution optical spectra. This work also reports the spectroscopic stellar parameters, temperature, and microturbulent velocity, as well as the stellar surface gravity from a solution of excitation and ionization equilibria of Fe I and Fe II lines under the assumption of local thermodynamic equilibrium. The abundance analysis reveals that HD 8270, HD 13551, and HD 22589 have metallicities of $[Fe/H] = -0.43, -0.28,$ and -0.12 , respectively. It was found that the abundances of iron group and α -elements follow the abundance pattern of a disk population. The heavy-element abundance patterns of the three stars show enhancements by a factor of 4–8 with respect to the Sun. The abundances of the *s*-process elements are discussed and compared with other barium giants and dwarfs through diagrams involving the indices $[hs/ls]$ and $[s/Fe]$. The metallicity distribution of barium giants and dwarfs is also discussed.

Key words: stars: abundances — stars: chemically peculiar — stars: general

1. INTRODUCTION

In 1974 H. Bond (Bond 1974) discovered, using objective-prism plates, a very interesting group of peculiar stars with spectral type G. The main peculiarity of these stars was the strength of Sr II $\lambda\lambda 4077$ –4215 and Ba II $\lambda 4554$, as well as the strength of the G band of CH. Further abundance studies (Snedden & Bond 1976; Luck & Bond 1982, 1991; Smith et al. 1993, hereafter SCL93; North et al. 1994) showed that these stars display anomalously strong heavy-element ($Z > 30$) abundances and also an overabundance of carbon with respect to the Sun ($[C/Fe] \approx +0.4$ to $+1.2$). In addition, the studies mentioned above confirmed the low luminosities of the stars through their high gravities. Thus, it was soon recognized that both CH subgiants and barium giant stars share the same abundance pattern; they are chemically peculiar stars in which the large overabundance of carbon and elements heavier than $Z > 30$ are produced by slow neutron capture reactions.

In all the studies mentioned above, the authors investigated all the CH subgiants known at that time. The number of new CH subgiants or CH subgiant candidates has been slowly increasing as a result of either spectroscopic abundance surveys of F and G dwarfs (HR 107, Tomkin et al. 1989; HD 21922, 36667, 80218, 140324, and 220842, Reddy et al. 2003, hereafter R03) or an application of the maximum likelihood principle to a sample of barium stars selected from the Lu (1991) catalog (Gómez et al. 1997; Mennessier et al. 1997). In short, this algorithm allows us to separate a sample of barium stars into groups according to their mean absolute magnitudes, kinematics, and *z*-scale heights.

The main purpose of this paper is to verify whether the stars HD 8270, HD 13551, and HD 22589 listed in Gómez et al. (1997) and Mennessier et al. (1997) as barium dwarf candidates display spectral characteristics (high gravities and overabundances of carbon and elements with $Z > 30$) typical of these stars. In a

previous investigation in this line of work, Pereira & Junqueira (2003) determined the barium dwarf nature of the candidate stars HD 50264 and HD 87080. For this investigation, high-resolution spectroscopy is used in order to obtain the stellar parameters and abundance patterns of the three stars currently being studied for comparison with those of other CH subgiant and barium stars already studied. The metallicity distribution between barium giants and dwarfs is also discussed.

2. OBSERVATIONS

The high-resolution spectra of HD 8270, HD 13551, and HD 22589 analyzed in this work were obtained at the 1.52 m ESO telescope of La Silla, Chile, on 2000 August 12 and October 15 and 16, respectively, with the FEROS echelle spectrograph (Kaufer et al. 1999). The FEROS spectral resolving power is $R = 48,000$, corresponding to 2.2 pixels of $15 \mu\text{m}$, and the wavelength coverage is from 4000 to 9200 Å. The nominal signal-to-noise ratio (S/N) was evaluated by the measurement of the rms flux fluctuation in selected continuum windows, and a typical value achieved was 150–200 after 2700–3600 s of integration time.

3. ANALYSIS AND RESULTS

3.1. Line Selection, Measurement, and Oscillator Strengths

The spectra of HD 8270, HD 13551, and HD 22589 show many atomic absorption lines of Fe I and Fe II, as well as transitions due to C I, Na I, Mg I, Si I, Ca I, Sc II, Ti I, Cr I, Ni I, Zn I, Cu I, Y II, Ba II, La II, Ce II, Nd II, and Eu II. The lines chosen are sufficiently unblended to yield reliable abundances. The selected lines are listed in Table 1 for Fe I and Fe II and in Table 2 for the other elements. Table 1 shows the Fe I and Fe II lines employed in the analysis; it also shows the lower excitation potential, χ , of the transitions, the *gf*-values, and the measured equivalent widths. The latter were obtained by fitting Gaussian profiles to the observed ones. The *gf*-values for the Fe I and Fe II lines in Table 1 were taken from Lambert et al. (1996). For the other elements, the sources of the *gf*-values are given in Table 2.

¹ Based on observations made with the 1.52 m telescope at the European Southern Observatory (La Silla, Chile) under agreement with Observatório Nacional (Brazil).

TABLE 1
OBSERVED Fe I AND Fe II LINES

ELEMENT	λ	χ	$\log gf$	EQUIVALENT WIDTHS (mÅ)		
				HD 8270	HD 13551	HD 22589
Fe I	5123.72	1.01	-3.070	...	94	...
	5133.69	4.18	0.201	108	119	...
	5150.84	0.99	-3.000	81	84	134
	5151.91	1.01	-3.320	69	72	...
	5159.06	4.28	-0.650	42	46	...
	5162.27	4.18	0.079	98	105	...
	5194.94	1.56	-2.090	91	95	...
	5198.71	2.22	-2.140	67	73	...
	5232.94	2.94	-0.080	157	159	...
	5242.49	3.63	-0.970	60	72	...
	5250.21	0.12	-4.920	32	36	77
	5253.03	2.28	-3.790	...	...	19
	5281.79	3.04	-0.830	96	97	139
	5288.52	3.69	-1.510	28	28	...
	5302.31	3.28	-0.740	90	103	133
	5307.36	1.61	-2.970	57	59	...
	5315.05	4.37	-1.400	...	17	37
	5321.11	4.43	-1.190	18	...	49
	5322.04	2.28	-2.840	30	29	67
	5339.93	3.27	-0.680	93	103	...
	5341.02	1.61	-1.950	109	119	...
	5364.87	4.45	0.230	82	92	...
	5367.47	4.42	0.439	...	106	...
	5369.96	4.37	0.540	103	...	...
	5371.49	0.96	-1.650	159	167	...
	5373.71	4.47	-0.710	34	38	65
	5389.48	4.42	-0.250	...	68	...
	5393.17	3.24	-0.720	90	113	133
	5400.50	4.37	-0.100	77	88	...
	5405.77	0.99	-1.850	131	130	...
	5410.91	4.47	0.400	90	100	...
	5417.03	4.42	-1.530	11	...	...
	5434.52	1.01	-2.120	116	118	...
	5441.34	4.31	-1.580	...	16	...
	5445.04	4.39	0.041	82	74	111
	5446.92	0.99	-1.910	132	135	...
	5506.78	0.99	-2.800	85	92	...
	5522.45	4.21	-1.400	16	...	46
	5554.90	4.55	-0.380	56	65	...
	5560.21	4.43	-1.040	26	29	53
	5563.60	4.19	-0.840	48	47	...
	5567.39	2.61	-2.560	31	...	...
	5569.62	3.42	-0.490	94	98	135
	5572.84	3.40	-0.280	116	...	...
	5576.09	3.43	-0.850	76	78	112
	5584.77	3.57	-2.170	14	...	...
	5624.02	4.39	-1.330	...	...	39
	5633.95	4.99	-0.120	38	44	...
	5635.82	4.26	-1.740	11	...	...
	5638.26	4.22	-0.720	47	53	82
	5686.53	4.55	-0.450	46	...	80
	5691.50	4.30	-1.370	17	...	44
	5705.47	4.30	-1.360	14	...	40
	5717.83	4.28	-0.979	32	35	67
	5731.76	4.26	-1.150	30	31	60
	5762.99	4.21	-0.410	73	...	...
	5806.73	4.61	-0.900	27	...	57
	5814.81	4.28	-1.820	...	...	24
	5852.22	4.55	-1.180	15	20	42
	5883.82	3.96	-1.210	37	...	67

TABLE 1—*Continued*

ELEMENT	λ	χ	$\log gf$	EQUIVALENT WIDTHS (mÅ)		
				HD 8270	HD 13551	HD 22589
	5916.25	2.45	−2.990	22	19	59
	5934.65	3.93	−1.020	44	46	78
	6016.66	3.55	−1.670	41	43	...
	6020.17	4.61	−0.210	...	81	...
	6024.06	4.55	−0.060	68	76	103
	6027.05	4.08	−1.090	40	38	68
	6056.01	4.73	−0.400	42	45	74
	6065.48	2.61	−1.530	78	84	...
	6079.01	4.65	−0.970	22	22	46
	6093.64	4.61	−1.350	12	...	33
	6096.66	3.98	−1.780	11	...	37
	6136.61	2.45	−1.400	96	98	...
	6137.69	2.59	−1.400	92	101	141
	6151.62	2.18	−3.290	18	19	...
	6157.73	4.08	−1.110	33	...	70
	6165.36	4.14	−1.470	18	21	48
	6170.51	4.79	−0.380	46	46	77
	6173.34	2.22	−2.880	33	...	73
	6187.99	3.94	−1.570	21	20	52
	6191.56	2.43	−1.420	91	96	...
	6200.31	2.60	−2.440	40	38	78
	6213.43	2.22	−2.480	51	53	...
	6230.72	2.56	−1.280	100	103	147
	6252.56	2.40	−1.720	83	82	...
	6265.13	2.18	−2.550	51	52	93
	6322.69	2.59	−2.430	40	42	83
	6380.74	4.19	−1.320	26	...	55
	6393.60	2.43	−1.430	91	90	...
	6411.65	3.65	−0.660	84	85	124
	6419.95	4.73	−0.090	53	60	87
	6421.35	2.28	−2.010	78	82	128
	6430.85	2.18	−2.010	77	84	...
	6469.19	4.83	−0.620	26	29	61
	6592.91	2.72	−1.470	...	79	...
	6593.87	2.44	−2.420	45	49	88
	6597.56	4.79	−0.920	16	20	47
	6609.11	2.56	−2.690	29	28	73
	6750.15	2.42	−2.620	33	41	80
	6752.71	4.64	−1.200	16	...	36
	6810.26	4.61	−0.990	16	28	53
	6820.37	4.64	−1.170	13	...	42
	6841.34	4.61	−0.600	34	42	...
	6858.15	4.61	−0.930	27	30	59
	7130.92	4.22	−0.700	50	62	...
	7132.99	4.08	−1.610	17	18	44
Fe II	4993.35	2.81	−3.670	32	35	49
	5132.66	2.81	−4.000	19	21	43
	5197.56	3.23	−2.250	78	92	...
	5234.62	3.22	−2.240	78	86	94
	5276.00	3.20	−1.910	105	...	...
	5284.10	2.89	−3.010	51	62	68
	5325.56	3.22	−3.170	41	40	54
	5414.05	3.22	−3.620	18	24	40
	5534.83	3.25	−2.770	56	69	76
	5991.37	3.15	−3.560	22	...	45
	6084.10	3.20	−3.800	14	21	30
	6149.25	3.89	−2.720	25	37	46
	6247.55	3.89	−2.340	50	60	63
	6416.92	3.89	−2.680	33	38	49
	6432.68	2.89	−3.580	31	38	51

TABLE 2
OTHER LINES STUDIED

λ (Å)	SPECIES	χ (eV)	gf	REFERENCES	EQUIVALENT WIDTHS (mÅ)		
					HD 8270	HD 13551	HD 22589
5380.317.....	C I	7.68	2.042e-02	1	51	37	...
7113.180.....	C I	8.64	1.117e-01	1	24	41	39
7115.190.....	C I	8.64	1.175e-01	1	21	39	36
7116.990.....	C I	8.64	8.318e-01	1	20	39	21
7119.660.....	C I	8.64	4.500e-01	1	15	28	18
6154.200.....	Na I	2.10	2.951e-02	2	16	20	40
6160.800.....	Na I	2.10	5.888e-02	2	23	42	56
8717.833.....	Mg I	5.91	1.956e-01	3	42	45	78
8736.040.....	Mg I	5.94	4.571e-01	3	96	103	123
6696.020.....	Al I	5.91	4.786e-02	2	...	...	37
6698.670.....	Al I	5.91	2.400e-02	2	...	...	23
5708.740.....	Si I	4.95	3.236e-02	4	56	...	...
5772.139.....	Si I	5.08	1.560e-02	4	29	...	58
5793.080.....	Si I	4.93	8.600e-03	4	25	36	57
5948.540.....	Si I	5.08	6.025e-02	4	56	71	...
6131.580.....	Si I	5.62	2.065e-02	4	...	...	29
6161.300.....	Ca I	2.52	9.333e-02	5	36	40	65
6166.440.....	Ca I	2.52	7.244e-02	6	39	...	71
6169.040.....	Ca I	2.52	1.585e-01	7	57	63	93
6169.560.....	Ca I	2.53	3.311e-01	7	78	73	109
6455.600.....	Ca I	2.51	4.571e-02	6	26	28	56
6471.660.....	Ca I	2.51	2.061e-01	6	59	68	92
5641.001.....	Sc II	1.50	1.528e-01	4	26	47	61
5657.896.....	Sc II	1.51	4.881e-01	4	54	69	83
5684.202.....	Sc II	1.51	1.411e-01	4	27	35	54
6245.620.....	Sc II	1.51	9.550e-02	8	22	27	49
6604.600.....	Sc II	1.36	4.900e-02	9	23	28	48
5783.073.....	Cr I	3.32	2.671e-01	4	10	...	27
5783.875.....	Cr I	3.32	4.427e-01	4	12	...	44
6330.097.....	Cr I	0.94	4.954e-04	10	5	...	30
6327.600.....	Ni I	1.68	7.709e-04	11	11	...	46
6482.800.....	Ni I	1.94	2.344e-02	11	12	19	48
6586.330.....	Ni I	1.95	1.549e-03	11	14	...	45
6767.770.....	Ni I	1.83	6.761e-03	11	44	46	...
4722.160.....	Zn I	4.01	4.074e-01	12	53	62	88
4810.530.....	Zn I	4.06	6.761e-01	12	59	60	75
4883.680.....	Y II	1.08	1.175e+00	13	88	...	...
4900.120.....	Y II	1.03	1.169e+00	13	82	96	104
5087.430.....	Y II	1.08	5.395e-01	4	72	87	100
5119.120.....	Y II	0.99	8.433e-02	4	36	43	72
5200.410.....	Y II	0.99	2.692e-01	13	67	81	87
5205.720.....	Y II	0.99	4.571e-01	13	75	...	...
5402.780.....	Y II	1.84	4.197e-01	4	35	...	59
5544.620.....	Y II	1.74	2.070e-01	4	20	27	43
4208.980.....	Zr II	0.71	3.467e-01	14	66	...	...
4317.300.....	Zr II	0.71	4.169e-02	14	37	...	...
6496.900.....	Ba II	0.60	4.170e-01	5	151	172	170
4662.510.....	La II	0.00	7.244e-02	15	22	...	...
4748.730.....	La II	0.93	3.467e-01	15	14	...	...
6320.430.....	La II	0.17	5.284e-02	15	...	...	22
6390.480.....	La II	0.32	2.750e-02	16	8	13	20
4562.360.....	Ce II	0.48	2.138e+00	17	43	...	...
4628.160.....	Ce II	0.52	1.819e+00	17	40	...	...
5187.457.....	Ce II	1.21	2.994e+00	4	19	23	21
5274.236.....	Ce II	1.28	2.450e+00	4	22	29	33
5330.580.....	Ce II	0.87	7.383e-01	4	11	...	21
5472.300.....	Ce II	1.25	1.656e+00	4	...	15	12

TABLE 2—*Continued*

λ (Å)	SPECIES	χ (eV)	gf	REFERENCES	EQUIVALENT WIDTHS (mÅ)		
					HD 8270	HD 13551	HD 22589
5212.360.....	Nd II	0.20	1.995e-01	18	17	...	...
5234.190.....	Nd II	0.55	3.467e-01	19	16	...	...
5249.580.....	Nd II	0.98	1.202e+00	19	27	...	...
5293.160.....	Nd II	0.82	6.310e-01	19	26	...	...
5311.460.....	Nd II	0.98	7.063e-01	4	...	7	13
5319.810.....	Nd II	0.55	4.467e-01	19	26	...	...
5485.710.....	Nd II	1.26	1.247e+00	19	...	...	11
5740.880.....	Nd II	1.16	7.674e-01	19	...	...	8

REFERENCES.—(1) Lambert et al. 1982; (2) Lambert & Luck 1978; (3) Wiese et al. 1969; (4) Luck & Bond 1985; (5) Wiese & Martin 1980; (6) Smith et al. 1986; (7) Drake & Smith 1991; (8) Youssef & Amer 1989; (9) Lawler & Dakin 1989; (10) Kurucz 1994a, 1994b; (11) Martin et al. 1988; (12) Biémont & Godefroid 1980; (13) Hannaford et al. 1982; (14) Biémont et al. 1981; (15) Mishenina & Kovtyukh 2001; (16) Smith et al. 1996; (17) Gratton & Sneden 1994; (18) Corliss & Bozman 1962 (modified Sneden et al. 1996); (19) Ward et al. 1984, 1985 (modified Sneden et al. 1996).

3.2. Determination of the Atmospheric Parameters

The determination of stellar atmospheric parameters, effective temperature (T_{eff}), surface gravity ($\log g$), microturbulence (ξ), and $[\text{Fe}/\text{H}]$ (throughout, the notation $[\text{X}/\text{H}] = \log[N(\text{X})/N(\text{H})]_{*} - \log[N(\text{X})/N(\text{H})]_{\odot}$ is used) are prerequisites for the determination of photospheric abundance. The gravity was determined by forcing Fe I and Fe II to yield the same iron abundance at the selected effective temperature. The microturbulent velocity was determined by forcing the abundance determined from individual Fe I lines to show no dependence on equivalent width. The solution for the excitation equilibrium was set to zero slope for the relation between the Fe I abundances and the excitation potential of the measured lines. The solution thus found is unique, depending only on a set of Fe I and Fe II lines and the atmospheric model employed, and yields as a by-product the metallicity of the star $[\text{Fe}/\text{H}]$. The atmospheric parameters were determined in the local thermodynamic equilibrium (LTE) model atmospheres of Kurucz (1993) using the spectral analysis code MOOG (Sneden 1973). The final adopted atmospheric parameters are given in Table 3. Figures 1–3 show the diagrams corresponding to the abundances of the individual Fe I lines plotted against the lower excitation potential and the reduced line strength following the best-fit parameters for HD 8270, HD 13551, and HD 22589, respectively.

The internal errors in our adopted effective temperatures (T_{eff}) and microturbulent velocity (ξ) can be determined from the uncertainty in the slope of the Fe I abundance versus excitation potential and Fe I versus reduced equivalent width (W_{λ}/λ) relations. These quantities are given in Table 3. The standard deviation in $\log g$ was set by changing this parameter around the adopted solution until the difference between the Fe I and Fe II

mean abundance differed by exactly 1 standard deviation of the $[\text{Fe}/\text{H}]$ mean value.

3.3. Abundance Analysis

The abundances of chemical elements in HD 8270, HD 13551, and HD 22589 are determined with LTE model atmosphere techniques described previously. In brief, equivalent widths are calculated by integration through a model atmosphere and are compared with the observed equivalent widths. The calculation is repeated, changing the abundance of the element in question, until a match is achieved. The line synthesis code MOOG was

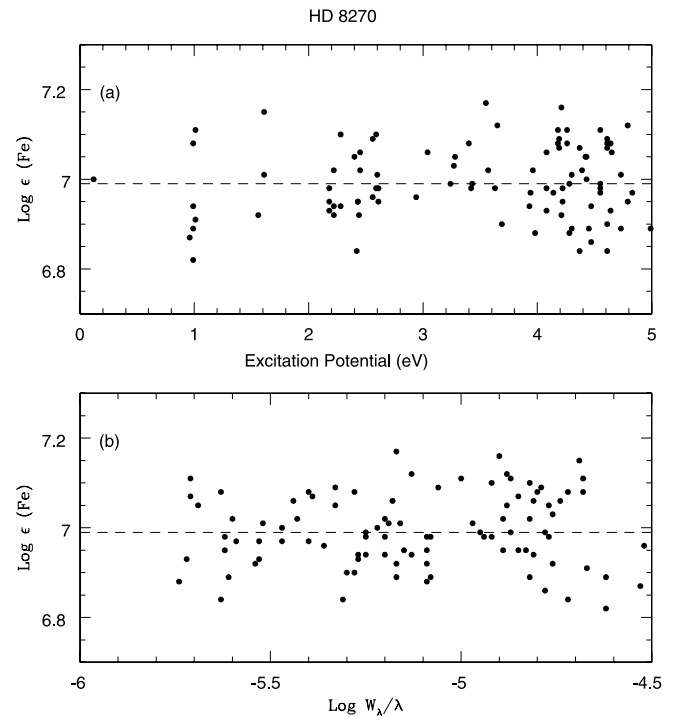


FIG. 1.—(a) Abundance of Fe by number $\epsilon(\text{Fe})$ vs. excitation potential and (b) abundance of Fe by number $\epsilon(\text{Fe})$ vs. $\log(W_{\lambda}/\lambda)$ for the 99 lines of Fe I employed in the determination of T_{eff} , ξ , and $\epsilon(\text{Fe})$ in HD 8270. The calculation was done with the adopted model $T_{\text{eff}} = 6100$ K, $\log g = 4.2$, $\epsilon(\text{Fe}) = 6.99$, and $\xi = 1.4$ km s $^{-1}$. The absence of any trend of the individual abundance with excitation potential and microturbulence shows that a correct value for T_{eff} and microturbulence velocity was chosen.

TABLE 3
ATMOSPHERIC AND PHYSICAL PARAMETERS OF HD 8270,
HD 13551, AND HD 22589

PARAMETER	VALUE		
	HD 8270	HD 13551	HD 22589
T_{eff} (K).....	6100 $\pm$ 200	6400 $\pm$ 100	5600 $\pm$ 200
$\log g$ (dex).....	4.2 $\pm$ 0.4	4.4 $\pm$ 0.4	3.8 $\pm$ 0.2
$[\text{Fe}/\text{H}]$ (dex).....	-0.53 $\pm$ 0.09	-0.28 $\pm$ 0.12	-0.16 $\pm$ 0.06
ξ (km s $^{-1}$).....	1.4 $\pm$ 0.4	1.6 $\pm$ 0.4	1.4 $\pm$ 0.3

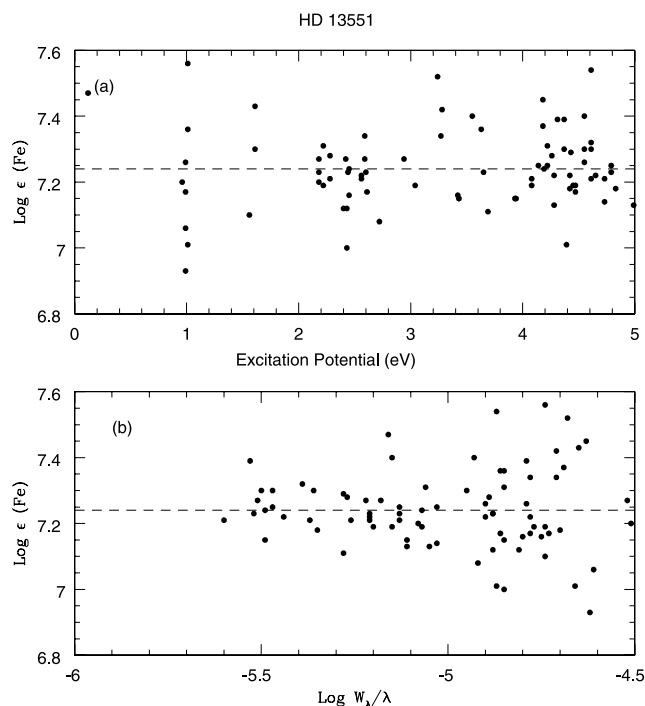


FIG. 2.—(a) Abundance of Fe by number $\epsilon(\text{Fe})$ vs. excitation potential and (b) abundance of Fe by number $\epsilon(\text{Fe})$ vs. $\log(W_\lambda/\lambda)$ for the 81 lines of Fe I employed in the determination of T_{eff} , ξ , and $\epsilon(\text{Fe})$ in HD 13551. The calculation was done with the adopted model $T_{\text{eff}} = 6400$ K, $\log g = 4.4$, $\epsilon(\text{Fe}) = 7.24$, and $\xi = 1.6$ km s $^{-1}$. The absence of any trend of the individual abundance with excitation potential and microturbulence shows that a correct value for T_{eff} and microturbulence velocity was chosen.

used to carry out the calculations. The results are given in Table 4, which also lists the number of lines employed for each species, n , and the standard deviation. The abundances of Mn I and Eu II are based on lines $\lambda 6021$ and $\lambda 6645$, and Cu I is based on lines $\lambda 5782$ for HD 8270 and HD 22589 and $\lambda 5015$ for HD 13551. The abundances for these elements were obtained through a spectrum synthesis technique. The synthetic spectra were broadened to match the instrumental profile. All fine-structure and hyperfine-structure components for the lines of Cu I, Mn I, and Eu II were explicitly included. The gf -values for Cu I are from Koch & Richter (1968), those for Mn I are from Booth et al. (1983), and those for Eu II are from Koch & Edvardsson (2002). Table 4 also shows the abundances obtained for these elements.

3.4. Abundance Uncertainties

The uncertainties in the derived abundances for the program stars are dominated by three main sources: the stellar parameters, the gf -values, and the equivalent width measurements. The uncertainties due to errors in the stellar atmospheric parameters were computed by changing these parameters by their standard errors, given in Table 3, and calculating the changes incurred in the element abundances. The results of these calculations are displayed in columns (2)–(6) of Tables 5–7.

The uncertainties due to the gf -values of the Fe I and Fe II lines (about 0.05 dex), elements up to Ni (0.02–0.10), and the heavy elements (± 0.2) were discussed in Smith et al. (1995). We thus refer to that paper for a detailed discussion of these uncertainties.

Errors in the equivalent widths are essentially set by the S/N and resolution of the spectra. The uncertainties due to equivalent width measurements were computed from an expression provided by Cayrel (1988). Applied to our spectra, with $R \approx 50,000$ and

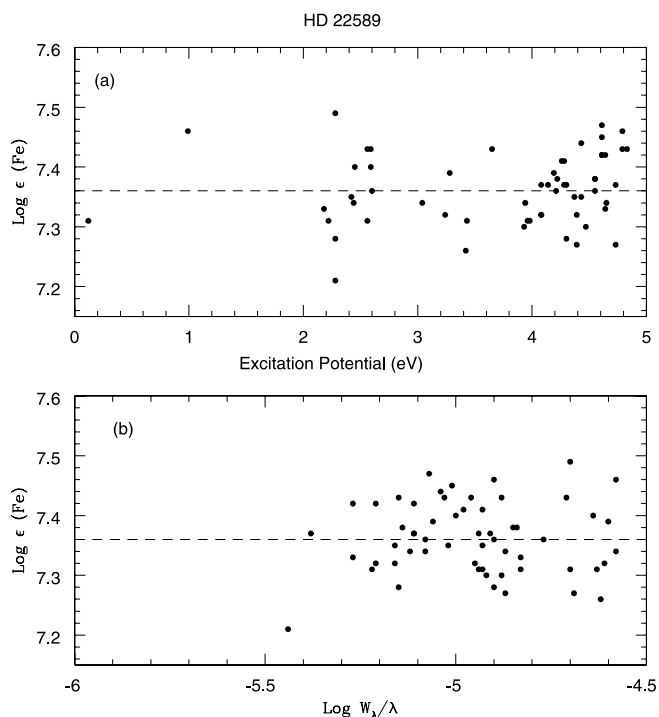


FIG. 3.—(a) Abundance of Fe by number $\epsilon(\text{Fe})$ vs. excitation potential and (b) abundance of Fe by number $\epsilon(\text{Fe})$ vs. $\log(W_\lambda/\lambda)$ for the 58 lines of Fe I employed in the determination of T_{eff} , ξ , and $\epsilon(\text{Fe})$ in HD 22589. The calculation was done with the adopted model $T_{\text{eff}} = 5600$ K, $\log g = 3.8$, $\epsilon(\text{Fe}) = 7.36$, and $\xi = 1.4$ km s $^{-1}$. The absence of any trend of the individual abundance with excitation potential and microturbulence shows that a correct value for T_{eff} and microturbulence velocity was chosen.

a typical S/N of 150, the expected uncertainties in the equivalent widths are about 2–3 mÅ. These error estimates have been applied to the stellar W_λ , and the corresponding changes in the element abundances are listed in column (6) of Tables 5–7.

Under the assumption that the errors are independent, they can be combined quadratically so that

$$\sigma = \left(\sum_{i=1}^N \sigma_i^2 \right)^{1/2}.$$

Combining the uncertainties as if they are independent is not purely accurate, as some of the uncertainties are correlated. While doing a full correlation analysis may not be necessary for this paper, it is noted that a change of +100 K in the temperature increases $\log g$ by +0.6 dex.

The final uncertainty of the abundances by number is found by quadratically combining the uncertainties due to atmospheric parameters, W_λ , and gf . This final uncertainty is given in column (7) of Tables 5–7. Column (8) gives the observed abundance dispersion among the lines for those elements with more than three available lines. These internal errors in the abundances are due to errors in the W_λ measurements, gf -values, microturbulence, T_{eff} , $\log g$, and non-LTE effects, which may account for the different lines.

Tables 5–7 show that neutral elements are sometimes sensitive to temperature variations, while singly ionized elements are sensitive to variations in $\log g$. The error introduced by the microturbulence is quite significant for elements whose abundance is based on stronger lines, such as barium. The uncertainties due to equivalent width errors affect only the abundances of elements, which are based on weak lines such as chromium and

TABLE 4
ABUNDANCES IN THE SCALE $\log \epsilon(\text{H}) = 12.0$ AND THE NOTATION $[\text{X}/\text{Fe}]$

SPECIES	HD 8270			HD 13551			HD 22589		
	n	$\log \epsilon$	$[\text{X}/\text{Fe}]$	n	$\log \epsilon$	$[\text{X}/\text{Fe}]$	n	$\log \epsilon$	$[\text{X}/\text{Fe}]$
Fe I.....	99	6.99 ± 0.09	...	81	7.24 ± 0.12	...	58	7.36 ± 0.06	...
Fe II.....	15	6.99 ± 0.12	...	14	7.26 ± 0.07	...	14	7.34 ± 0.09	...
C I.....	5	8.70 ± 0.20	+0.71	5	8.70 ± 0.10	+0.46	4	9.00 ± 0.12	+0.64
Na I.....	2	5.88	+0.08	2	6.26 ± 0.09	+0.21	2	6.21	+0.02
Mg I.....	2	7.12	+0.07	2	7.21	-0.09	2	7.39	-0.03
Al I.....	...	...	...	...	...	...	2	6.18	-0.13
Si I.....	4	7.32 ± 0.11	+0.32	2	7.66	+0.39	3	7.63 ± 0.06	+0.24
Ca I.....	6	5.92 ± 0.09	+0.12	5	6.11 ± 0.07	+0.03	6	6.21 ± 0.10	+0.04
Sc II.....	5	2.49 ± 0.08	-0.15	5	3.01 ± 0.09	+0.12	5	2.95 ± 0.05	-0.06
Cr I.....	3	5.21 ± 0.13	+0.07	...	...	...	2	5.63	+0.12
Mn I.....	1	4.89	-0.07	1	4.89	-0.22	1	5.39	+0.16
Ni I.....	4	5.67 ± 0.12	-0.05	2	6.04	+0.07	3	6.13 ± 0.09	-0.02
Cu I.....	1	3.61	-0.07	1	3.96	+0.03	1	4.11	+0.06
Zn I.....	2	4.11	+0.04	2	4.36	+0.04	2	4.50	+0.06
Y II.....	8	2.46 ± 0.15	+0.75	5	2.86 ± 0.14	+0.90	6	2.80 ± 0.13	+0.72
Zr II.....	2	2.78	+0.71	...	...	...	...	...	...
Ba II.....	1	2.77	+1.17	1	3.23	+1.38	1	2.72	+0.75
La II.....	4	1.39 ± 0.16	+0.75	1	2.12	+1.23	2	1.57	+0.56
Ce II.....	5	1.73 ± 0.22	+0.68	3	2.22 ± 0.16	+0.92	4	1.71 ± 0.23	+0.29
Nd II.....	5	1.77 ± 0.12	+0.80	1	1.75	+0.53	3	1.41 ± 0.06	+0.07
Eu II.....	1	0.21 ± 0.13	+0.19	...	...	...	1	0.61	+0.26

TABLE 5
ABUNDANCE UNCERTAINTIES OF HD 8270

Species (1)	ΔT_{eff} (2)	$\Delta \log g$ (3)	$\Delta \xi$ (4)	$\Delta[\text{Fe}/\text{H}]$ (5)	ΔW_{λ} (6)	$(\sum \sigma^2)^{1/2}$ (7)	σ_{obs} (8)
Fe I.....	+0.14	-0.04	+0.19	+0.06	+0.06	0.25	0.11
Fe II.....	+0.01	-0.14	+0.09	+0.02	+0.08	0.19	0.09
C I.....	-0.11	-0.14	+0.01	-0.07	+0.08	0.21	0.21
Na I.....	-0.08	+0.01	0.00	+0.03	+0.08	0.12	...
Mg I.....	+0.05	+0.06	+0.02	+0.02	+0.04	0.09	...
Si I.....	0.00	-0.05	-0.05	-0.05	-0.02	0.09	0.07
Ca I.....	+0.11	+0.02	+0.05	+0.04	+0.05	0.14	0.08
Sc II.....	+0.05	-0.15	+0.03	+0.04	+0.06	0.18	0.09
Cr I.....	+0.12	+0.01	0.00	+0.06	+0.15	0.20	0.09
Mn I.....	+0.10	0.00	0.00	+0.06	+0.05	0.13	...
Ni I.....	+0.16	0.00	+0.02	+0.07	+0.09	0.20	0.12
Cu I.....	+0.12	0.00	-0.10	+0.07	+0.05	0.18	...
Zn I.....	+0.10	-0.03	+0.14	+0.05	+0.07	0.19	...
Y II.....	-0.07	-0.13	+0.19	+0.06	+0.07	0.25	0.10
Zr II.....	-0.01	-0.23	+0.09	-0.03	-0.02	0.25	...
Ba II.....	-0.14	0.00	+0.13	+0.10	+0.03	0.22	...
La II.....	+0.08	-0.16	+0.03	-0.01	+0.10	0.21	0.07
Ce II.....	+0.08	-0.14	+0.09	+0.07	+0.09	0.21	0.10
Nd II.....	+0.09	-0.14	+0.05	+0.07	+0.08	0.20	0.14
Eu II.....	+0.10	-0.10	0.00	+0.06	+0.05	0.16	...

NOTES.—Col. (1): Species. Col. (2): Variation of the abundance ratios caused by a +200 K variation in T_{eff} . Cols. (3)–(6): Variations due to $\Delta \log g = +0.4$, $\Delta \xi = +0.4 \text{ km s}^{-1}$, $\Delta[\text{Fe}/\text{H}] = 0.09$, and $\Delta W_{\lambda} = +0.3 \text{ m\AA}$, respectively. The equivalent width and gf uncertainties are discussed in the text. Col. (7): Compounded rms uncertainty of cols. (2)–(6). Col. (8): Observed abundance dispersion of those elements with more than three available lines.

TABLE 6
ABUNDANCE UNCERTAINTIES OF HD 13551

Species (1)	ΔT_{eff} (2)	$\Delta \log g$ (3)	$\Delta \xi$ (4)	$\Delta[\text{Fe}/\text{H}]$ (5)	ΔW_{λ} (6)	$(\sum \sigma^2)^{1/2}$ (7)	σ_{obs} (8)
Fe I.....	+0.07	+0.04	-0.10	+0.02	+0.06	0.14	0.12
Fe II.....	-0.01	-0.16	-0.07	-0.03	+0.06	0.19	0.07
C I.....	-0.06	-0.13	-0.01	+0.03	+0.06	0.15	0.10
Na I.....	+0.05	+0.02	+0.01	0.00	+0.06	0.08	...
Mg I.....	+0.02	+0.05	+0.02	+0.01	+0.03	0.07	...
Si I.....	+0.04	+0.01	+0.04	+0.01	+0.05	0.08	...
Ca I.....	+0.06	+0.03	+0.05	+0.01	+0.06	0.10	0.07
Sc II.....	+0.01	-0.16	-0.04	-0.05	+0.05	0.18	0.09
Mn I.....	+0.11	+0.01	0.00	+0.05	+0.04	0.13	...
Ni I.....	+0.08	0.00	-0.05	0.00	+0.06	0.11	...
Cu I.....	+0.12	-0.01	-0.09	+0.06	+0.05	0.17	...
Zn I.....	+0.05	-0.03	-0.11	-0.01	-0.04	0.13	...
Y II.....	+0.02	-0.14	-0.15	-0.04	+0.06	0.22	0.14
Ba II.....	+0.05	-0.01	-0.20	-0.07	+0.03	0.22	...
La II.....	+0.03	-0.16	-0.02	-0.07	+0.10	0.20	...
Ce II.....	+0.02	-0.15	-0.03	-0.06	+0.08	0.18	0.16
Nd II.....	+0.03	-0.15	-0.01	-0.07	+0.17	0.24	...

NOTES.—Col. (1): Species. Col. (2): Variation of the abundance ratios caused by a +100 K variation in T_{eff} . Cols. (3)–(6): Variations due to $\Delta \log g = +0.4$, $\Delta \xi = +0.4 \text{ km s}^{-1}$, $\Delta[\text{Fe}/\text{H}] = 0.14$, and $\Delta W_{\lambda} = +0.3 \text{ mÅ}$, respectively. The equivalent width and gf uncertainties are discussed in the text. Col. (7): Compounded rms uncertainty of cols. (2)–(6). Col. (8): Observed abundance dispersion of those elements with more than three available lines.

TABLE 7
ABUNDANCE UNCERTAINTIES OF HD 22589

Species (1)	ΔT_{eff} (2)	$\Delta \log g$ (3)	$\Delta \xi$ (4)	$\Delta[\text{Fe}/\text{H}]$ (5)	ΔW_{λ} (6)	$(\sum \sigma^2)^{1/2}$ (7)	σ_{obs} (8)
Fe I.....	+0.16	-0.01	-0.11	0.00	+0.05	0.20	0.12
Fe II.....	-0.05	+0.08	-0.08	+0.02	+0.07	0.14	0.09
C I.....	-0.16	+0.04	-0.01	+0.01	+0.07	0.18	0.13
Na I.....	+0.11	-0.01	-0.02	0.00	+0.04	0.12	...
Mg I.....	+0.05	-0.05	-0.03	-0.01	+0.03	0.08	...
Al I.....	+0.10	0.00	-0.01	0.00	+0.06	0.12	...
Si I.....	+0.07	+0.02	-0.02	0.00	+0.06	0.10	0.07
Ca I.....	+0.15	-0.02	-0.09	0.00	+0.05	0.18	0.10
Sc II.....	+0.01	+0.08	-0.06	-0.03	+0.06	0.12	0.06
Ti I.....	+0.19	-0.01	-0.02	0.00	+0.07	0.20	...
Cr I.....	+0.16	0.00	-0.03	-0.01	+0.06	0.17	0.07
Mn I.....	+0.12	+0.02	0.00	+0.04	+0.04	0.13	...
Ni I.....	+0.18	+0.02	-0.05	-0.01	+0.05	0.19	0.10
Cu I.....	+0.11	+0.01	-0.08	+0.05	+0.04	0.15	...
Zn I.....	+0.05	+0.03	-0.16	-0.02	+0.07	0.19	...
Y II.....	+0.03	+0.07	-0.17	-0.03	+0.07	0.20	0.13
Ba II.....	+0.08	+0.02	-0.16	+0.05	+0.03	0.19	...
La II.....	+0.04	+0.08	-0.02	+0.04	+0.08	0.13	...
Ce II.....	+0.03	-0.08	-0.03	+0.04	+0.08	0.13	0.23
Nd II.....	+0.04	+0.09	-0.01	+0.03	+0.15	0.18	0.06
Eu II.....	+0.09	-0.10	0.01	+0.07	+0.04	0.16	...

NOTES.—Col. (1): Species. Col. (2): Variation of the abundance ratios caused by a +200 K variation in T_{eff} . Cols. (3)–(6): Variations due to $\Delta \log g = +0.2$, $\Delta \xi = +0.3 \text{ km s}^{-1}$, $\Delta[\text{Fe}/\text{H}] = 0.06$, and $\Delta W_{\lambda} = +0.3 \text{ mÅ}$, respectively. The equivalent width and gf uncertainties are discussed in the text. Col. (7): Compounded rms uncertainty of cols. (2)–(6). Col. (8): Observed abundance dispersion of those elements with more than three available lines.

lanthanum in HD 8270, lanthanum and neodymium in HD 13551, and neodymium in HD 22589. For the elements analyzed via spectrum synthesis, the same technique was used: varying T_{eff} , $\log g$, and ξ , then independently computing the abundance changes introduced by the variation of the above atmospheric parameters. They are also included in Tables 5–7.

4. DISCUSSION

4.1. Spectral Type and Distance of HD 8270, HD 13551, and HD 22589

Previous spectroscopic observations of HD 8270 and HD 22589 were done by MacConnell et al. (1972) and Lu (1991) with photographic plates. Section 3.2 shows that the computed atmospheric parameters are similar to the ones found by those authors. HD 8270 was classified as F8 by MacConnell et al. (1972) and as F8 IV by Lu (1991), while HD 22589 was classified as G0 by MacConnell et al. (1972) and as G5 by Lu (1991). In addition, the determination of atmospheric parameters was in good agreement with the classification by Gómez et al. (1997) and Mennessier et al. (1997) of HD 8270 and HD 22589 as dwarf barium stars.

HD 13551 was classified by MacConnell et al. (1972) as a marginal barium star of spectral type G0, thus belonging to their Table 2, but of spectral type F6 by Lu (1991). Gómez et al. (1997) classified HD 13551 as a subgiant star, while Mennessier et al. (1997) listed it as a halo candidate.

Our spectroscopic gravities determined from ionization equilibria agree well with the distance determination given by astrometric data (Mennessier et al. 1997). The luminosities of all three stars were estimated from the relation

$$L = 4\pi\sigma R_*^2 T_{\text{eff}}^4, \quad (1)$$

using our derived T_{eff} and $\log g$. It is necessary to assume a mass for HD 8270, HD 13551, and HD 22589. The value $M_* = 0.8 M_\odot$ was adopted for all three stars by taking the turnoff masses for the 12 Gyr isochrones of Vandenberg (2000), given the metallicities of the stars analyzed in this study. Considering the gravity and the uncertainties obtained (Table 3), the luminosities of HD 8270, HD 13551, and HD 22589 are, re-

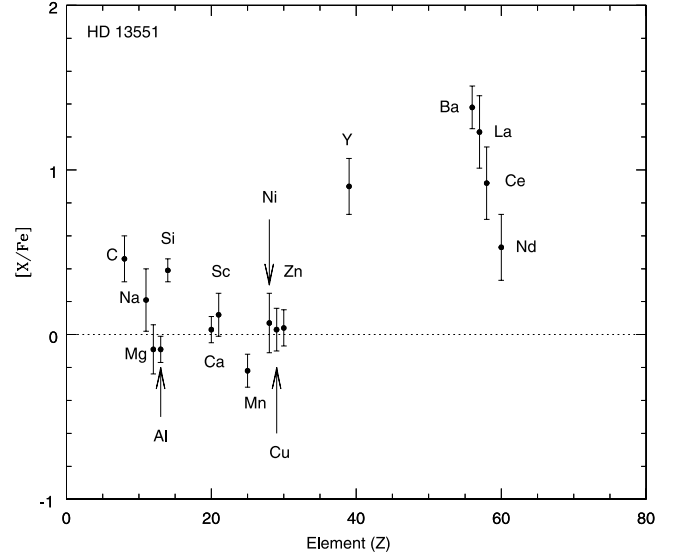


FIG. 5.—Abundance pattern of HD 13551. The error bars were taken from col. (7) of Table 6 and represent the combined errors for each element.

spectively, $L = 1.7^{+2.14}_{-0.79}$, $1.3^{+2.28}_{-0.82}$, and $3.1^{+2.5}_{-1.5} L_\odot$. This corresponds to $M_V = +4.06^{+0.11}_{-0.82}$, $+4.34^{+1.14}_{-1.06}$, and $+3.4^{+0.6}_{-0.7}$ for HD 8270, HD 13551, and HD 22589, respectively, assuming no bolometric correction and $M_{\text{bol}\odot} = +4.64$ (Schmidt-Kaler 1982). One then enters the observed $V = 9.2$ (HD 8270), $V = 9.4$ (HD 13551), and $V = 9.0$ (Lu 1991) to obtain $r = 107^{+6}_{-34}$, 103^{+72}_{-40} , and 132^{+42}_{-37} pc for HD 8270, HD 13551, and HD 22589, respectively. However, with the uncertainties of approximately 100–200 K and 0.4 dex in $\log g$ for HD 8270 and HD 13551, these limits easily accommodate the Mennessier et al. (1997) values of 78 and 118 pc, respectively. The positions of these stars in the H-R diagram are the same as those of the other sample analyzed by Luck & Bond (1991, hereafter LB91; see their Fig. 5). For HD 22589, there is no previous distance determination.

4.2. The Abundance Patterns of HD 8270, HD 13551, and HD 22589

Here the abundance patterns of HD 8270, HD 13551, and HD 22589 are compared with those of CH subgiants found in previous studies. Figures 4–6 show the abundance patterns of HD 8270, HD 13551, and HD 22589 analyzed in this work. These three stars display very similar abundance patterns to those of two other barium dwarfs previously analyzed, HD 50264 and HD 87080 (Pereira & Junqueira 2003).

The abundances of α -elements (Mg, Si, and Ca) of HD 8270, HD 13551, and HD 22589 are, respectively, +0.15, +0.11, and +0.10. The $[\alpha/\text{Fe}]$ mean given by $[\langle \text{Mg, Si, Ca} \rangle / \text{Fe}]$ for a sample of disk F and G dwarfs lies between 0.1 and 0.15 (R03). Similar values were found for the sample stars analyzed in SCL93 and LB91 of 0.14 (taking into consideration only silicon and calcium) and 0.18, respectively.

The iron group elements follow Fe. Nickel is basically solar, as are scandium and chromium. The abundance for the iron group is similar to other determinations for CH subgiant stars, 0.06 (SCL93) and 0.08 (LB91). Manganese abundance is also similar to these, which are similar to the value given by SCL93 of 0.09.

The rest of the elements within the iron group have small enhancements, as is the case for zinc (+0.04 for HD 8270 and

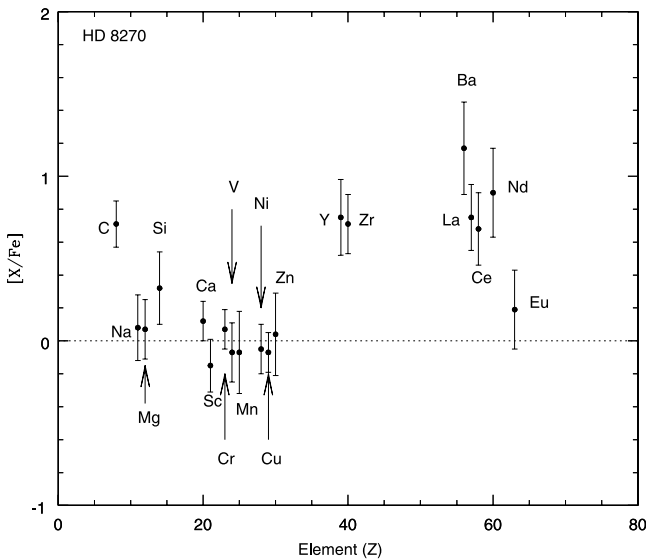


FIG. 4.—Abundance pattern of HD 8270. The error bars were taken from col. (7) of Table 5 and represent the combined errors for each element.

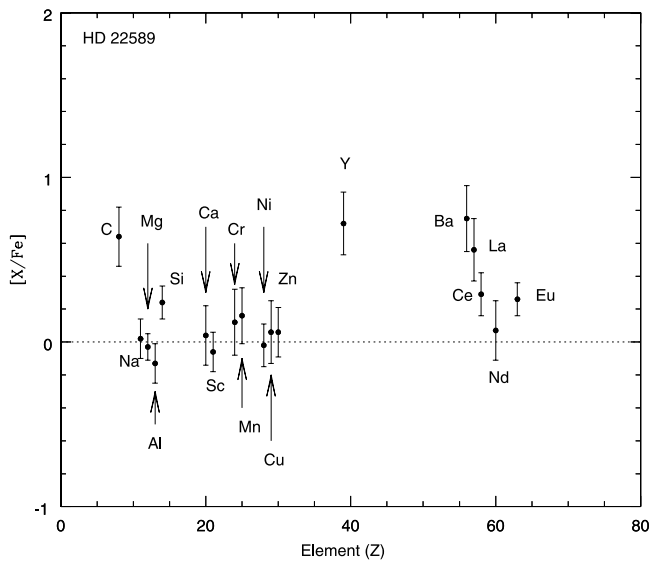


FIG. 6.—Abundance pattern of HD 22589. The error bars were taken from col. (7) of Table 7 and represent the combined errors for each element.

HD 13551 and +0.11 for HD 22589) and copper (+0.08 for HD 8270, +0.03 for HD 13551, and +0.12 for HD 22589). Copper and zinc were analyzed by SCL93 and LB91 with a mean of 0.00 ± 0.28 and 0.11 ± 0.12 , respectively. Copper and zinc were also analyzed in R03. The abundance ratios $[\text{Cu}/\text{Fe}]$ and $[\text{Zn}/\text{Fe}]$ are basically solar for a sample of disk stars at the same metallicity analyzed by R03. Sodium shows the same trend as for disk dwarf stars, with a mean of 0.15 for $[\text{Na}/\text{Fe}]$ for a -0.4 to -0.2 metallicity star (R03).

The abundance of europium for HD 8270 and HD 22589 lies slightly above the tendency seen in Figure 3 of Koch & Edvardsson (2002). In that plot, the $[\text{Eu}/\text{Fe}]$ ratio for the disk and halo dwarf stars is plotted versus the metallicity; the mean value of $[\text{Eu}/\text{Fe}]$ for the metallicity of HD 8270 ($[\text{Fe}/\text{H}] = -0.53$) is ≈ 0.2 , while for the metallicity of HD 22589 ($[\text{Fe}/\text{H}] = -0.16$) it is ≈ 0.1 .

Lithium abundances in barium dwarf and giant stars have already been investigated by Smith & Lambert (1986) and Lambert et al. (1993). They concluded that these two classes are Li-poor and that the absorption feature at $6707.8 \text{ \AA}$ might be Ce II. The stars analyzed in this work also confirm the trend seen in barium dwarf stars; no lithium line was detected around $6707 \text{ \AA}$.

Inspection of Figures 4–6 reveals that HD 8270, HD 13551, and HD 22589 show a general overabundance of elements heavier than Rb synthesized by neutron capture. The available abundances for low-metallicity dwarf stars (R03) indicate that the chemical evolution of the Galaxy cannot be responsible for the s -process overabundances observed in these stars.

The abundance patterns of HD 8270, HD 13551, and HD 22589 imply a mean heavy s -process $[\text{s}/\text{Fe}]$ element excess for Y, Zr, Ba, La, Ce, and Nd of 0.81, 0.99, and 0.48, respectively. Similar values are found among other studies already done for CH subgiants, such as SCL93, LB91, Pereira & Junqueira (2003), and North et al. (1994). Figure 7 shows $[\text{s}/\text{Fe}]$ versus $[\text{Fe}/\text{H}]$ for a sample of barium giants, CH subgiants, HD 50264, HD 87080, and the three stars analyzed in this work. The figure shows that the parameter $[\text{s}/\text{Fe}]$ is anticorrelated with metallicity. The positions of HD 8270, HD 13551, and HD 22589 analyzed in this work are the same as those of the other stars of the same class. As Clayton (1988) has shown, the clear significant trend of

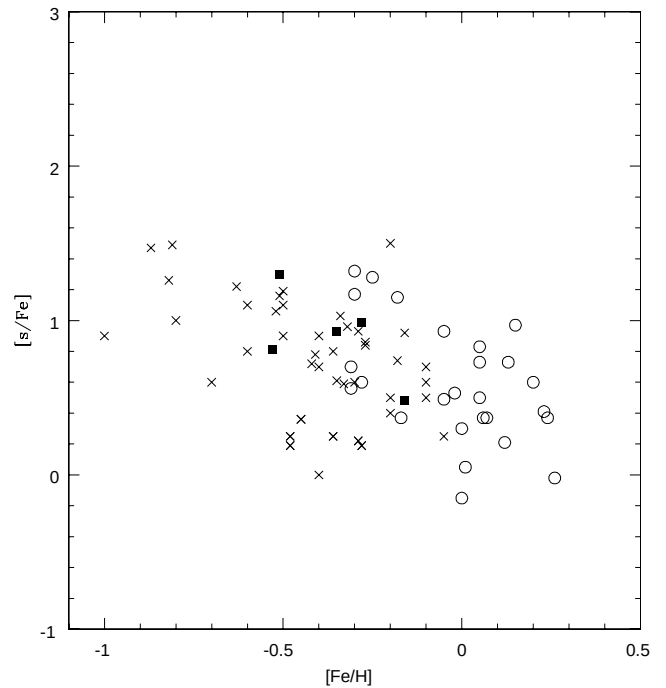


FIG. 7.—Diagram of the mean s -process elements given by the ratio $[\text{s}/\text{Fe}]$ as a function of the metallicity ($[\text{Fe}/\text{H}]$) for a sample of s -process enriched objects. Information is plotted for HD 8270, HD 13551, and HD 22589 (this work) and HD 50264 and HD 87080 (Pereira & Junqueira 2003; *filled squares*); CH subgiants and barium dwarfs (North et al. 1994; SCL93; Luck & Bond 1991; Tomkin et al. 1989; R03; *crosses*); and barium giants (Smith 1984; Zacs 1994; *open circles*).

increasing $[\text{hs}/\text{ls}]$ (as well as $[\text{s}/\text{Fe}]$) with decreasing metallicity is the result of the operation of the reaction $^{13}\text{C}(\alpha, n)^{16}\text{O}$ as a neutron source. As the $[\text{s}/\text{Fe}]$ index is anticorrelated with metallicity, the carbon abundance is also anticorrelated. Figure 8 shows $[\text{C}/\text{Fe}]$ versus $[\text{Fe}/\text{H}]$ for a sample of barium giants and dwarfs and CH subgiants.

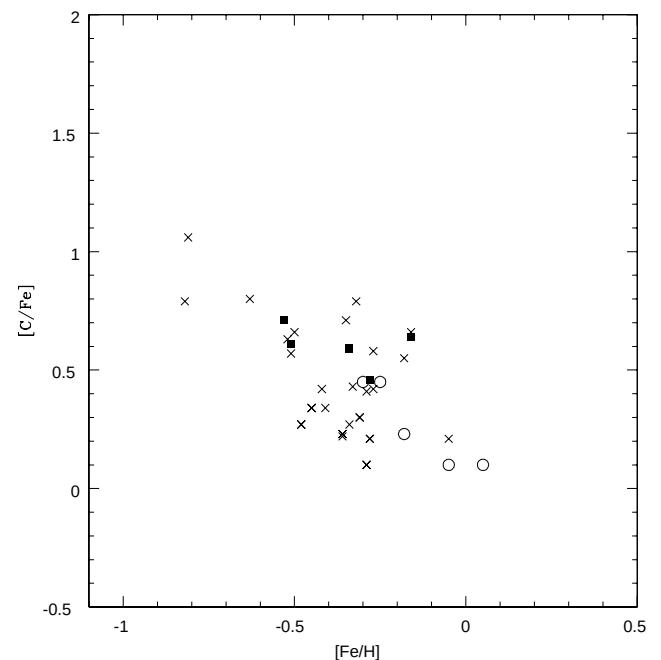


FIG. 8.—Diagram of $[\text{C}/\text{Fe}]$ ratio vs. metallicity. The symbols have the same meaning as in Fig. 7.

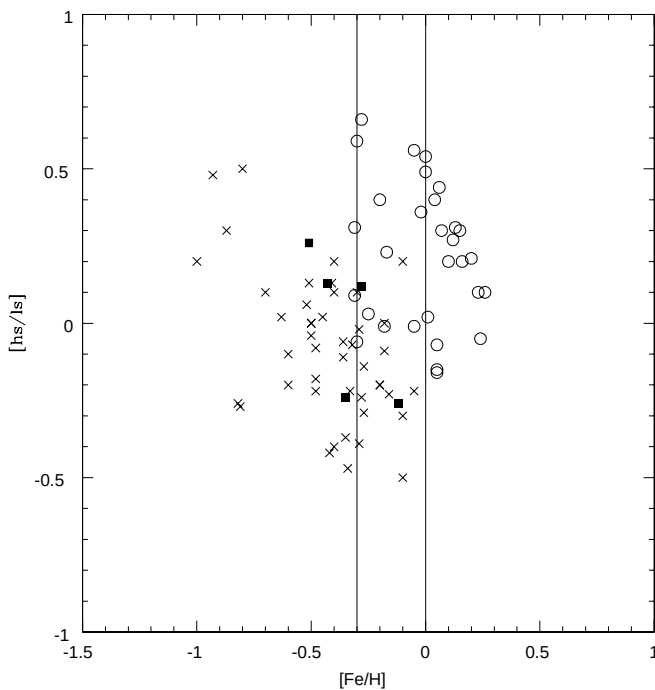


FIG. 9.—Diagram of the neutron exposure parameter estimator $[hs/ls]$ as a function of the metallicity ($[Fe/H]$) for a sample of s -process enriched objects. The symbols have the same meaning as in Fig. 7.

To summarize this section, the abundances of the three stars analyzed in this work display the same pattern as that of all CH subgiants analyzed so far; i.e., carbon and elements heavier than $Z > 30$ have overabundances with respect to the Sun, while the other elements have abundances that are basically solar.

4.3. Comparison between Barium Giants and Dwarfs

Figure 9 shows the neutron exposure index $[hs/ls]$ versus metallicity ($[Fe/H]$) for a sample of barium giants and dwarfs. This index is a good measure of the determination of the neutron exposure parameter (τ_0) of Cowley & Downes (1980). We can see that in this range of metallicity there is little dependence of the s -process overabundances on $[Fe/H]$ for barium dwarfs and giants. In addition, we also see that both classes of stars display a wide range of $[hs/ls]$ values, from -0.5 to 0.5 . This observational fact has been interpreted by several authors (LB91, SCL93) as an indication that barium dwarf stars do become barium giant stars. If barium dwarfs evolve through a subgiant phase into barium giants, then these classes of peculiar stars should show considerable similarity of composition. In fact, the neutron exposures determined so far for both classes of stars present similar values. For two subgiant CH stars, Krishnaswamy & Sneden (1985) found $\tau_0 = 0.2$; for HD 50264 and HD 87080, a revised value of τ_0 previously determined by Pereira & Junqueira (2003) gives $\tau_0 \approx 0.4$; HD 8270, HD 13351, and HD 2259 give (using data from this work) $\tau_0 = 0.3$ – 0.4 and (using data from SCL93) $\tau_0 = 0.4$. For a sample of barium giants, Smith (1984) found $\tau_0 = 0.3$ – 0.4 , and Zacs (1994) found $\tau_0 = 0.4$ – 0.5 . Another argument in favor

of the evolution of a barium dwarf into a barium giant comes from the study of North et al. (2000). In that study the authors searched for radial velocity variations in a sample of barium giants and dwarfs and concluded that from the “orbital” point of view the most massive dwarfs should become barium giants.

Figure 9 also shows that barium dwarfs do not have metallicities higher than $[Fe/H] \geq 0.0$. This metallicity distribution has already been observed by LB91. In Figure 6 of their study the authors noted in a histogram of $[Fe/H]$ ratios for barium giants and dwarfs that the latter represents a group of stars of intermediate to old age disk population, while the former represents a younger population. If barium dwarfs evolve into barium giants, where are the barium dwarfs with metallicities in the range 0.0 – 0.3 or the real disk CH subgiants? We may also formulate this question the other way around: Where are the barium giants with metallicities smaller than $[Fe/H] \leq -0.5$? Regarding the second question, these kinds of stars were first recognized by LB91 based on objective-prism plates. They called these stars metal-deficient barium stars, the Population II analogs of classical barium stars (those with metallicities in the range 0.0 – 0.3). Junqueira & Pereira (2001) added another star to this family, HD 206983, with $[Fe/H] = -1.43$. However, Jorissen (2003) later claimed that one of the stars analyzed by Junqueira & Pereira (2001), HD 104340, might not belong to the group of metal-deficient barium stars, since it did not exhibit radial velocity variation. Jorissen (2003) lists several candidates for metal-deficient barium stars yet to be investigated.

The first question (where are the disk CH subgiants or those with metallicities in the range $0.0 < [Fe/H] < 0.3$?) may have already been partially answered. There exists a class of rapidly rotating K dwarf and barium-enriched stars known as wind-accretion-induced, rapidly rotating (WIRring) stars (Jeffries & Smalley 1996). In respect to the question raised above, the star analyzed by those authors is a solar-metallicity disk star. Another star type that deserves a comment is the dwarf carbon star, or dC star (Green & Margon 1994). As the halo dwarf carbon stars would evolve into CH stars, the disk dC stars would evolve into CH subgiants and then into classical barium stars (Green & Margon 1994). CH subgiants at near solar metallicities are yet to be found.

5. CONCLUSIONS

An abundance analysis, employing high-resolution optical spectra, of three CH subgiant stars reveals that they are substantially enriched in the heavy s -process elements. The abundance patterns for the iron peak elements, α group as well as sodium and aluminum, follow the disk abundance. With respect to s -process nucleosynthesis, HD 8270, HD 13551, and HD 22589 display values that fit well the observed behavior of other CH subgiant stars (Pereira & Junqueira 2003; SCL93; LB91).

The metallicities of the CH subgiants analyzed in this work indicate that they should also belong to an old-disk population, as do all the CH subgiants analyzed so far. It still remains to find CH subgiants at near solar metallicity, the precursor of the classical barium stars. Clearly, barium stars that have still not been identified as giants or dwarfs and their respective metallicities need to be spectroscopically observed.

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